

Shobhit Singh



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EDUCATION

Iowa State University, College of Engineering Ames, Iowa
Bachelor of Science in Computer Engineering

Fall 2024- Fall 2026

Bennett University Gr. Noida, India (**Transferred to ISU**)
Bachelor of Technology in Computer Science Engineering

Fall 2022 – Spring 2024

EXPERIENCE

Horse's Mouth Noida, India

Mar 2023 – Aug 2023

Operations and Technical Content Intern

- Partnered directly with CEO on mentor recruitment, business strategy, and workflow redesign, boosting website traffic **6x** and increasing user registrations from ~10 in 4 months to **10+ per month**.
- **Led a 6-member content team**, launching AI-driven SEO campaigns and producing high-engagement posts across Quora, Reddit, Twitter, and Instagram.

PROJECTS

PerchGuard: Motion-Triggered Bird Feeder Monitor (Seeed XIAO ESP32-S3)

- Built motion detection pipeline with ROI masking; captures 10-sec video/audio clips on trigger and securely sends to phone.
- Added power-efficient design (deep sleep + wake on motion) and optional MJPEG live view.

CANary: Automotive CAN-Bus Telemetry & Edge Alerts (Seeed XIAO ESP32-S3)

- Prototyped OBD-II/CAN capture; decoded key PIDs and published health snapshots via BLE/Wi-Fi.
- Integrated mic/camera for acoustic and visual correlation during fault events.

Autonomous Wi-Fi Optimization & Router Placement Robot (Arduino Uno + Tiva C)

- Mapped Wi-Fi strength across rooms and visualized placement suggestions via Python GUI.

ML-Based Multifunctional Audio Tool: (Python, Streamlit)

- Implemented noise cancellation, transcription, and audio generation in a lightweight web interface.

RESEARCH

Undergraduate Researcher: Quantum Networking Protocols (ISU)

Fall 2025 – Present

- Investigating link-layer and transport abstractions for near-term quantum networks.
- Modelling protocol state machines and analysing reliability/latency trade-offs.

Chaos Theory for Image Encryption

- Developed a novel chaos function for image encryption.

Advances in Trusted Execution Environments for Blockchain Security

- Reviewed TEEs for securing Hyperledger Fabric, with insights into Web3 and blockchain security.

Comparative Analysis of Pre- & Post-Quantum Encryption Algorithms

- Researched and compared encryption algorithms in the context of quantum-safe cryptography.

LEADERSHIP

General Secretary: Tau Sigma Transfer Honors Society (ISU)

Fall 2025 – Present

General Secretary: ACM Student Chapter (Bennett Univ.)

Aug 2023 – May 2024

Chairperson: Quantum Computing Club (Bennett Univ.)

Aug 2023 – May 2024

SKILLS

Languages: C++, Java, Python, HTML, CSS, JavaScript, Embedded C

Embedded Systems: ESP-IDF/Arduino, FreeRTOS, SPI/I²C/UART, BLE/Wi-Fi, RTSP/MJPEG, basic OpenCV/TFLite-Micro

Networking & Security: CAN (ISO-11898), OBD-II (SAE J1979), APIs/HTTP, Cryptography basics

Data/ML: NumPy, pandas, TensorFlow (intro), Streamlit

Tools: Git, Linux, VS Code, MATLAB (coursework)

Quantum: Qiskit, protocol modeling